

WITCHCLIFFE, Australia

WMO: 956410

Lat: 34.0281S

Lon: 115.1042E

Elev: 266

StdP: 14.56

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.1	41.3	36.1	31.5	46.7	37.6	33.4	48.1	23.8	60.4	22.3	60.0	0.5	150	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.0	88.4	67.0	84.5	66.2	80.7	65.0	69.4	81.6	67.9	78.8	66.6	76.5	11.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	98.9	70.5	64.8	92.8	69.1	63.4	88.1	68.8	33.7	82.2	32.3	79.0	31.4	76.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.1	20.4	18.2	34.6	97.2	2.3	2.6	33.0	99.1	31.7	100.6	30.4	102.1	28.8	104.0		
			34.8	72.2	1.7	1.6	33.6	73.3	32.5	74.3	31.6	75.2	30.3	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	67.3	68.0	65.8	61.4	57.6	55.0	53.6	53.9	54.6	57.2	61.3	64.7
	DBStd	6.69	4.74	5.10	5.08	4.34	3.56	3.59	3.71	3.34	3.46	3.92	4.67	4.90
	HDD50	20	0	0	0	0	1	2	9	5	3	0	0	0
	HDD65	2199	25	22	50	124	229	299	354	343	311	246	131	65
	CDD50	3670	536	505	490	343	238	153	120	127	142	222	340	454
	CDD65	373	96	107	75	16	1	0	0	0	0	3	20	55
	CDH74	2980	733	831	585	118	11	0	0	0	1	34	210	457
	CDH80	952	240	281	203	19	0	0	0	0	0	3	59	147
(14) Wind	WSAvg	8.4	11.2	10.5	8.6	6.9	6.4	6.9	7.5	7.7	8.4	8.6	9.4	9.5
(15) Precipitation	PrecAvg	40.9	0.6	0.5	1.3	2.4	5.8	7.7	7.9	6.1	4.3	2.4	1.4	0.8
	PrecMax	51.5	2.5	2.2	4.4	5.7	9.8	11.5	14.8	8.7	8.2	5.0	4.1	2.3
	PrecMin	30.6	0.0	0.0	0.3	0.2	2.2	4.1	3.3	1.7	1.5	0.9	0.0	0.2
	PrecStd	5.6	0.7	0.5	0.9	1.5	1.9	2.3	2.6	1.4	1.7	1.1	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	92.4	93.2	82.8	75.2	67.8	66.1	67.5	71.3	78.4	86.9	91.6
		MCWB	68.1	68.8	68.0	63.6	59.6	58.7	59.7	58.9	58.0	63.5	63.9	66.2
	2%	DB	86.4	87.4	85.4	77.3	70.7	65.7	63.6	64.8	66.6	72.6	80.1	84.3
		MCWB	67.6	67.8	66.3	63.2	60.2	58.2	57.8	57.8	58.6	61.5	63.2	65.6
	5%	DB	81.6	83.2	79.9	73.5	67.9	64.1	62.0	62.9	64.1	68.3	74.7	79.0
		MCWB	65.7	66.3	65.1	62.8	60.0	57.4	56.5	56.8	57.5	59.3	61.8	64.6
	10%	DB	77.4	79.0	75.8	70.5	65.8	62.4	60.3	61.1	62.0	65.2	70.9	74.8
		MCWB	64.4	65.3	63.8	61.4	58.9	56.6	55.2	55.5	55.8	57.9	60.4	62.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	72.2	70.2	67.5	65.4	62.0	62.3	62.3	63.1	65.2	67.0	69.5
		MCDB	84.0	83.0	82.6	73.3	68.9	64.2	63.5	64.3	66.3	74.6	79.8	83.6
	2%	WB	69.0	69.4	68.2	65.7	63.2	60.6	59.6	60.0	60.4	62.6	64.8	67.0
		MCDB	81.0	81.6	77.3	71.9	67.0	63.5	62.1	62.9	64.4	69.6	74.1	79.6
	5%	WB	67.2	67.7	66.7	64.2	61.3	59.0	57.6	58.1	58.1	60.6	63.3	65.0
		MCDB	78.2	79.5	75.1	70.4	65.5	62.5	60.7	61.6	62.7	66.1	71.4	75.6
	10%	WB	65.5	66.3	65.3	62.7	59.9	57.5	55.9	56.2	56.5	59.0	61.7	63.5
		MCDB	75.5	76.4	73.6	68.6	64.3	61.2	59.4	60.2	61.0	63.8	69.5	72.6
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.0	19.2	17.9	16.5	13.6	12.7	13.3	13.9	15.5	17.9	18.7
		MCDBR	27.5	27.4	27.4	23.5	21.1	16.2	14.4	16.9	18.7	22.6	25.7	26.3
		MCWBR	11.1	10.8	11.2	11.3	12.5	11.0	9.9	10.6	11.7	12.3	11.7	10.9
	5% WB	MCDBR	23.0	22.6	21.1	18.4	16.0	12.9	12.1	13.7	15.6	18.4	21.5	22.3
		MCWBR	10.0	9.6	9.7	10.6	11.4	10.4	10.1	10.5	11.2	11.2	11.0	9.9
(40) Clear-Sky Solar Irradiance	taub		0.332	0.332	0.322	0.317	0.312	0.306	0.303	0.314	0.327	0.324	0.328	0.327
	taud		2.599	2.609	2.638	2.646	2.628	2.626	2.634	2.590	2.534	2.557	2.572	2.596
	Ebn at Noon		319	312	305	289	273	266	273	285	298	311	318	322
	Edh at Noon		33	31	28	25	23	21	22	26	31	33	33	33
(44) All-Sky Solar Radiation	RadAvg		2435	2167	1724	1231	894	715	770	1041	1404	1851	2226	2433
(45)	RadStd		126	93	101	69	58	46	36	57	75	103	121	119

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page